1 Navier Stokes Equations and Continuity equation 2D

In this section we present the equations which describe the dynamical behaviour of a flow assuming incompressibility, i.e. the air density ρ is a constant. Derivations and more background information on these equations can be found in **?**[Versteeg: An introduction to CFD]. For each cell in the filed, there are three momentum equations and one continuity equation. The former can be seen as the second law of Newton, i.e. $\dot{L} = \sum_i F_i$ and the latter as a stronger, local form of the law of conservation. In this document we will focus on two dimensions hence there are two momentum equations and one continuity equation for each cell in the field.

u-momentum equation

In this section we will spatially discretise the u-momentum equation using the finite volume method. We have the u-momentum equation

$$\rho\left(\underbrace{\frac{\partial u}{\partial t}}_{\text{variation}} + \underbrace{\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y}}_{\text{convection}}\right) = -\underbrace{\frac{\partial p}{\partial x}}_{\text{pressure gradient}} + \mu\left(\underbrace{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}}_{\text{diffusion}}\right) + \underbrace{S_{M_u}}_{\text{external force}} + \underbrace{T_{M_u}}_{\text{turbulence}}$$

When spatially discretising using the finite volume method, we integrate each term over one control volume ΔV. After doing so, the above can be written as:

$$\rho\frac{\partial u}{\partial t}\Delta V + a_P u_P = -(p_e - p_w)\delta_{y_v} + a_E u_E + a_W u_W + a_N u_N + a_S u_S + \bar{S}_{M_u}\Delta V + \bar{T}_{M_u}\Delta V$$

with $\Delta V = \delta_{x_u}\delta_{y_v}$ and

$$a_E = \max\{-F_e, D_e - \frac{F_e}{2}, 0\}, \quad a_W = \max\{F_w, D_w + \frac{F_w}{2}, 0\}$$
$$a_N = \max\{-F_n, D_n - \frac{F_n}{2}, 0\}, \quad a_S = \max\{F_s, D_s + \frac{F_s}{2}, 0\}$$

$a_P = a_E + a_W + a_N + a_S + F_e - F_w + F_n - F_s$. We furthermore have that

$$F_e = \rho u_e A_e, \quad F_w = \rho u_w A_w, \quad D_e = \frac{\mu A_e}{\delta_{x_{PE}}}, \quad D_w = \frac{\mu A_w}{\delta_{x_{PW}}}$$
$$F_n = \rho v_n A_n, \quad F_s = \rho v_s A_s, \quad D_n = \frac{\mu A_n}{\delta_{y_{PN}}}, \quad D_s = \frac{\mu A_s}{\delta_{y_{PS}}}$$

Note that for the pressure gradient we used the approximation $\frac{\partial p}{\partial x} = \frac{p_e - p_w}{\delta_{x_u}}$. The above equations are also valid for the v-momentum equation though due to the fact that the latter will be discretised over another control volume, the indices will be different. In Fig.1 we have the three control volumes

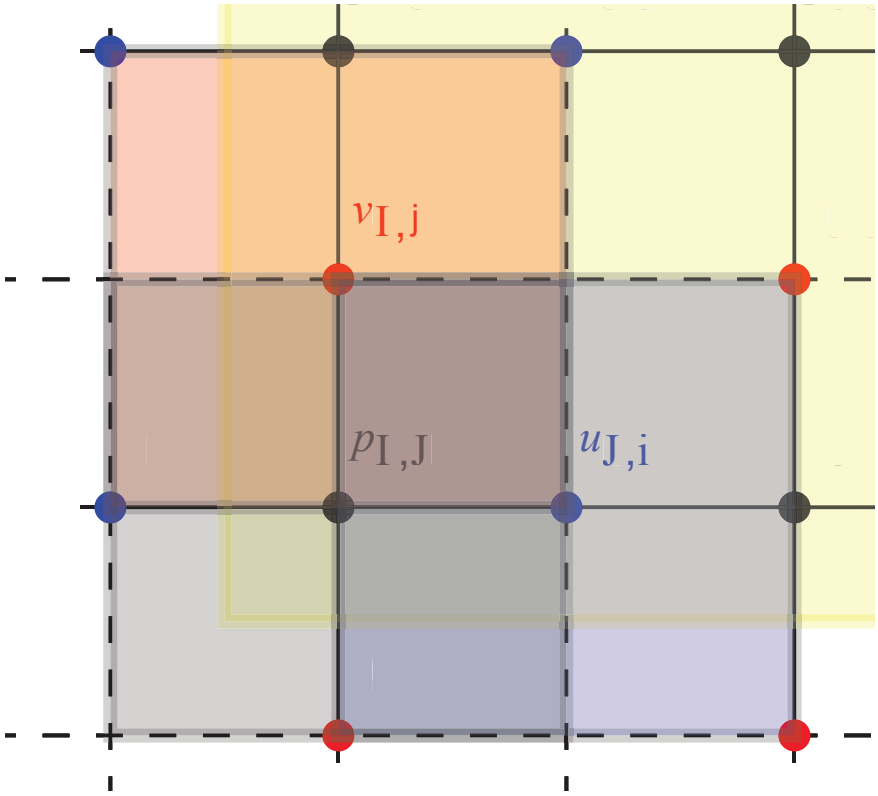


Figure 1: Reference frame used to discretise the u-momentum (blue), v-momentum (red) and continuity equation (black).

which will be use to discretise the u-momentum, v-momentum and continuity equation. According to the grid we can formulate a general equation for the u-momentum:

$$\left(S^x_{i,J} - \rho\frac{\partial u_{i,J}}{\partial t}\right)\underbrace{(x_{I+1,J} - x_{I,J})}_{\Delta x_{I,I+1}}\underbrace{(y_{i,j} - y_{i,j-1})}_{\Delta y_{j-1,j}} = a^{P_x}_{i,J}u_{i,J} - a^{E_x}_{i+1,J}u_{i+1,J} - a^{W_x}_{i-1,J}u_{i-1,J} - \cdots$$
$$\cdots - a^{N_x}_{i,J+1}u_{i,J+1} - a^{S_x}_{i,J-1}u_{i,J-1} + (p_{I+1,J} - p_{I,J})\underbrace{(y_{i,j} - y_{i,j-1})}_{\Delta y_{j-1,j}}$$

Define the range $i = [3, N_x - 1], i^x_0 = 3, I \in [2, N_x - 1], I^x_0 = 2, J \in [2, N_y - 1], J^x_0 = 2, j \in [3, N_y], j^x_0 = 3$ and

$$u = \begin{pmatrix} u_{i,J} & u_{i,J+1} & \cdots & u_{i,N_y-1} & u_{i+1,J} & u_{i+1,J+1} & \cdots & u_{N_x-1,J} & \cdots & u_{N_x-1,N_y-1}\end{pmatrix}^T \in \mathbb{R}^{(N_y-J_0)(N_x-i_0)}$$
$$p = \begin{pmatrix} p_{I,J} & p_{I,J+1} & \cdots & p_{I,N_y-1} & p_{I+1,J} & p_{I+1,J+1} & \cdots & p_{N_x-1,J} & \cdots & p_{N_x-1,N_y-1}\end{pmatrix}^T \in \mathbb{R}^{(N_y-J_0)(N_x-I_0)}$$

Furthermore we define the equation

$$V_x\left(S_x - \rho\frac{\partial u}{\partial t}\right) + u^0 + p_x^0 = A_x u + B_x p$$

Hence we have the following

$$S_x - \rho\frac{\partial u}{\partial t} = \begin{pmatrix} S^x_{i,J} - \rho\dfrac{\partial u_{i,J}}{\partial t} \\ S^x_{i,J+1} - \rho\dfrac{\partial u_{i,J+1}}{\partial t} \\ \vdots \\ S^x_{i,N_y-1} - \rho\dfrac{\partial u_{i,N_y-1}}{\partial t} \\ S^x_{i+1,J} - \rho\dfrac{\partial u_{i+1,J}}{\partial t} \\ S^x_{i+1,J+1} - \rho\dfrac{\partial u_{i+1,J+1}}{\partial t} \\ \vdots \\ S^x_{N_x-1,J} - \rho\dfrac{\partial u_{N_x-1,J}}{\partial t} \\ \vdots \\ S^x_{N_x-1,N_y-1} - \rho\dfrac{\partial u_{N_x-1,N_y-1}}{\partial t} \end{pmatrix}, \quad u^0 = \begin{pmatrix} a^{W_x}u_{i-1,J} + a^{S_x}u_{i,J-1} \\ a^{W_x}u_{i-1,J+1} \\ \vdots \\ a^{W_x}u_{i-1,N_y-1} + a^{N_x}u_{i,N_y} \\ a^{S_x}u_{i+1,J-1} \\ 0 \\ \vdots \\ a^{S_x}u_{N_x-1,J-1} + a^{E_x}u_{N_x,J} \\ \vdots \\ a^{N_x}u_{N_x-1,N_y} + a^{E_x}u_{N_x,N_y-1} \end{pmatrix}, \quad p_x^0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$V_x = \mathtt{diag}(\Delta x_{I,I+1}\Delta y_{j-1,j}, \Delta x_{I,I+1}\Delta y_{j,j+1}, \ldots, \Delta x_{I,I+1}\Delta y_{N_y-1,N_y}, \Delta x_{I+1,I+2}\Delta y_{j-1,j}, \\ \Delta x_{I+1,I+2}\Delta y_{j,j+1}, \ldots, \Delta x_{N_x-2,N_x-1}\Delta y_{j-1,j}, \ldots, \Delta x_{N_x-2,N_x-1}\Delta y_{N_y-1,N_y})$$

$$A^x_{i,J} = \begin{pmatrix} a^{P_x}_{i,J} & -a^{N_x}_{i,J+1} & 0_{N_y-J_0-2} & -a^{E_x}_{i+1,J} & 0_{N_y-J_0-1} & 0_{(N_y-J_0)(N_x-i_0-2)} \end{pmatrix}$$

$$A^x_{i,J+1} = \begin{pmatrix} -a^{S_x}_{i,J} & a^{P_x}_{i,J+1} & -a^{N_x}_{i,J+2} & 0_{N_y-J_0-3} & 0 & -a^{E_x}_{i+1,J+1} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)((N_x-i_0-2)} \end{pmatrix}$$

$$\vdots$$

$$A^x_{i,N_y-1} = \begin{pmatrix} 0_{N_y-J_0-2} & -a^{S_x}_{i,N_y-2} & a^{P_x}_{i,N_y-1} & 0_{N_y-J_0-1} & -a^{E_x}_{i+1,N_y-1} & 0_{(N_y-J_0)(N_x-i_0-2)} \end{pmatrix}$$

$$A^x_{i+1,J} = \begin{pmatrix} -a^{W_x}_{i,J} & 0_{N_y-J_0-1} & a^{P_x}_{i+1,J} & -a^{N_x}_{i+1,J+1} & 0_{N_y-J_0-2} & -a^{E_x}_{i+2,J} & 0_{N_y-J_0-1} & 0_{(N_y-J_0)(N_x-i_0-2} \end{pmatrix}$$

$$A^x_{i+1,J+1} = \begin{pmatrix} 0 & -a^{W_x}_{i,J+1} & 0_{N_y-J_0-2} & -a^{S_x}_{i+1,J} & a^{P_x}_{i+1,J+1} & -a^{N_x}_{i+1,J+2} & 0_{N_y-J_0-3} & 0 & -a^{E_x}_{i+2,J+1} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)(N_x-i_0-3)} \end{pmatrix}$$

$$\vdots$$

$$A^x_{N_x-1,J} = \begin{pmatrix} 0_{(N_y-J_0)(N_x-i_0-2)} & -a^{W_x}_{N_x-2,J} & 0_{N_y-J_0-1} & a^{P_x}_{N_x-1,J} & -a^{N_x}_{N_x-1,J+1} & 0_{N_y-J_0-2} \end{pmatrix}$$

$$\vdots$$

$$A^x_{N_x-1,N_y-1} = \begin{pmatrix} 0_{(N_y-J_0)(N_x-i_0-2)} & 0_{(N_y-J_0-1)} & -a^{W_x}_{N_x-2,N_y-1} & 0_{N_y-J_0-2} & -a^{S_x}_{N_x-1,N_y-2} & a^{P_x}_{N_x-1,N_y-1} \end{pmatrix}$$

and

$$A_x = \begin{bmatrix} A^x_{i,J} \\ A^x_{i,J+1} \\ \vdots \\ A^x_{i,N_y-1} \\ A^x_{i+1,J} \\ A^x_{i+1,J+1} \\ \vdots \\ A^x_{N_x-1,J} \\ \vdots \\ A^x_{N_x-1,N_y-1} \end{bmatrix} \in \mathbb{R}^{(N_y-J_0)(N_x-i_0)\times(N_y-J_0)(N_x-i_0)} \tag{1}$$

At last we have

$$\begin{aligned}
B^x_{i,J} &= \begin{pmatrix} -\Delta y_{j-1,j} & 0_{N_y-J_0-1} & \Delta y_{j-1,j} & 0_{N_y-J_0-1} & 0_{(N_y-J_0)(N_x-I_0-2)} \end{pmatrix} \\
B^x_{i,J+1} &= \begin{pmatrix} 0 & -\Delta y_{j,j+1} & 0_{N_y-J_0-2} & 0 & \Delta y_{j,j+1} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)(N_x-I_0-2)} \end{pmatrix} \\
&\vdots \\
B^x_{i,N_y-1} &= \begin{pmatrix} 0_{N_y-J_0-1} & -\Delta y_{N_y-1,N_y} & 0_{N_y-J_0-1} & \Delta y_{N_y-1,N_y} & 0_{(N_y-J_0)(N_x-I_0-2)} \end{pmatrix} \\
B^x_{i+1,J} &= \begin{pmatrix} 0_{N_y-J_0} & -\Delta y_{j-1,j} & 0_{N_y-J_0-1} & \Delta y_{j-1,j} & 0_{N_y-J_0-1} & 0_{(N_y-J_0)(N_x-I_0-3)} \end{pmatrix} \\
B^x_{i+1,J+1} &= \begin{pmatrix} 0_{N_y-J_0} & 0 & -\Delta y_{j,j+1} & 0_{N_y-J_0-2} & 0 & \Delta y_{j,j+1} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)(N_x-I_0-3)} \end{pmatrix} \\
&\vdots \\
B^x_{N_x-1,J} &= \begin{pmatrix} 0_{(N_y-J_0)(N_x-I_0-2)} & -\Delta y_{j-1,j} & 0_{N_y-J_0-1} & \Delta y_{j-1,j} & 0_{N_y-J_0-1} \end{pmatrix} \\
&\vdots \\
B^x_{N_x-1,N_y-1} &= \begin{pmatrix} 0_{(N_y-J_0)(N_x-I_0-2)} & 0_{N_y-J_0-1} & -\Delta y_{N_y-1,N_y} & 0_{N_y-J_0-1} & \Delta y_{N_y-1,N_y} \end{pmatrix}
\end{aligned}$$

so that we can construct

$$B_x = \begin{bmatrix} B^x_{i,J} \\ B^x_{i,J+1} \\ \vdots \\ B^x_{i,N_y-1} \\ B^x_{i+1,J} \\ B^x_{i+1,J+1} \\ \vdots \\ B^x_{N_x-1,J} \\ \vdots \\ B^x_{N_x-1,N_y-1} \end{bmatrix} \tag{2}$$

The coefficients are defined as:

$$a^{E_x}_{i+1,J} = \max\{-F^{e_x}_{I+1,J}, D^{e_x}_{I+1,J} - \frac{F^{e_x}_{I+1,J}}{2}, 0\}, \quad a^{W_x}_{i-1,J} = \max\{F^{w_x}_{I,J}, D^{w_x}_{I,J} + \frac{F^{w_x}_{I,J}}{2}, 0\}$$

$$a^{N_x}_{i,J+1} = \max\{-F^{n_x}_{i,j}, D^{n_x}_{i,j} - \frac{F^{n_x}_{i,j}}{2}, 0\}, \qquad a^{S_x}_{i,J-1} = \max\{F^{s_x}_{i,j-1}, D^{s_x}_{i,j-1} + \frac{F^{a_x}_{i,j-1}}{2}, 0\}$$

$a^{P_x}_{i,J} = a^{E_x}_{i+1,J} + a^{W_x}_{i-1,J} + a^{N_x}_{i,J+1} + a^{S_x}_{i,J-1} + F^{e_x}_{I+1,J} - F^{w_x}_{I,J} + F^{n_x}_{i,j} - F^{s_x}_{i,j-1}$. with

$$D^{e_x}_{I+1,J} = \frac{\mu \Delta y_{j,j+1}}{\Delta x_{i,i+1}}, \qquad D^{w_x}_{I,J} = \frac{\mu \Delta y_{j,j+1}}{\Delta x_{i-1,i}} \tag{3}$$

$$D^{n_x}_{i,j} = \frac{\mu \Delta x_{I,I+1}}{\Delta y_{J,J+1}}, \qquad D^{s_x}_{i,j-1} = \frac{\mu \Delta x_{I,I+1}}{\Delta y_{J-1,J}} \tag{4}$$

$$F^{e_x}_{I+1,J} = \frac{\rho \Delta y_{j,j+1}}{2}\left(u_{i+1,J} + u_{i,J}\right), \quad F^{w_x}_{I,J} = \frac{\rho \Delta y_{j,j+1}}{2}\left(u_{i-1,J} + u_{i,J}\right) \tag{5}$$

$$F^{n_x}_{i,j} = \frac{\rho \Delta x_{I,I+1}}{2}\left(v_{I,j} + v_{I+1,j}\right), \quad F^{s_x}_{i,j-1} = \frac{\rho \Delta x_{I,I+1}}{2}\left(v_{I,j-1} + v_{I+1,j-1}\right) \tag{6}$$

v-momentum equation

In this section we will spatially discretise the v-momentum equation using the finite volume method. The equation is defined as:

$$\rho\left(\frac{\partial v}{\partial t}+\frac{\partial vu}{\partial x}+\frac{\partial v^2}{\partial y}\right)=-\frac{\partial p}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)+S_{M_v}+T_{M_v}$$

The pressure gradient will be approximated using $\frac{\partial p}{\partial y}=\frac{p_n-p_s}{\delta_{yv}}$. Otherwise, the same equations hold as given in the previous subsection though the indices will not be the same due to the fact that we do not use the same control volume to discretise the equation. The general formulation is as follows.

$$\left(S^y_{I,j}-\rho\frac{\partial v_{I,j}}{\partial t}\right)\underbrace{(x_{i,j}-x_{i-1,j})}_{\Delta x_{i-1,i}}\underbrace{(y_{I,J+1}-y_{I,J})}_{\Delta y_{J,J+1}}=a^{P_y}_{I,j}v_{I,j}-a^{E_y}_{I+1,j}v_{I+1,j}-a^{W_y}_{I-1,j}v_{I-1,j}-\cdots$$
$$\cdots-a^{N_y}_{I,j+1}v_{I,j+1}-a^{S_y}_{I,j-1}v_{I,j-1}+(p_{I,J+1}-p_{I,J})\underbrace{(x_{i,j}-x_{i-1,j})}_{\Delta x_{i-1,i}}$$

Define

$$v=\begin{pmatrix}v_{I,j} & v_{I,j+1} & \cdots & v_{I,N_y-1} & v_{I+1,j} & v_{I+1,j+1} & \cdots & v_{N_x-1,j} & \cdots & v_{N_x-1,N_y-1}\end{pmatrix}^T\in\mathbb{R}^{(N_y-j_0)(N_x-I_0)}$$

with $i\in[3,N_x], i^y_0=3, I\in[2,N_x-1], I^y_0=2, J\in[2,N_y-1], J^y_0=2, j\in[3,N_y-1], j^y_0=3$. Define the equation

$$V_y\left(S_y-\rho\frac{\partial v}{\partial t}\right)+v^0+p^0_y=A_yv+B_yp$$

Hence we have the following

$$S_y - \rho\frac{\partial v}{\partial t} = \begin{pmatrix} S^y_{I,j} - \rho\dfrac{\partial v_{I,j}}{\partial t} \\ S^y_{I,j+1} - \rho\dfrac{\partial v_{I,j+1}}{\partial t} \\ \vdots \\ S^y_{I,N_y-1} - \rho\dfrac{\partial v_{I,N_y-1}}{\partial t} \\ S^y_{I+1,j} - \rho\dfrac{\partial v_{I+1,j}}{\partial t} \\ S^y_{I+1,j+1} - \rho\dfrac{\partial v_{I+1,j+1}}{\partial t} \\ \vdots \\ S^y_{N_x-1,j} - \rho\dfrac{\partial v_{N_x-1,j}}{\partial t} \\ \vdots \\ S^y_{N_x-1,N_y-1} - \rho\dfrac{\partial v_{N_x-1,N_y-1}}{\partial t} \end{pmatrix}, \quad v^0 = \begin{pmatrix} a^{W_y}v_{I-1,j} + a^{S_y}v_{I,j-1} \\ a^{W_y}v_{I-1,j+1} \\ \vdots \\ a^{W_y}v_{I-1,N_y-1} + a^{N_y}v_{I,N_y} \\ a^{S_y}v_{I+1,j-1} \\ 0 \\ \vdots \\ a^{S_y}v_{N_x-1,j-1} + a^{E_y}v_{N_x,j} \\ \vdots \\ a^{N_x}v_{N_x-1,N_y} + a^{E_y}v_{N_x,N_y-1} \end{pmatrix}, \quad p^0_y = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$V_y = \mathtt{diag}(\Delta x_{i-1,i}\Delta y_{J,J+1}, \Delta x_{i-1,i}\Delta y_{J+1,J+2}, \ldots, \Delta x_{i-1,i}\Delta y_{N_y-1,N_y}, \Delta x_{i,i+1}\Delta y_{J,J+1}, \\ \Delta x_{i,i+1}\Delta y_{J+1,J+2}, \ldots, \Delta x_{N_x-1,N_x}\Delta y_{J,J+1}, \ldots, \Delta x_{N_x-1,N_x}\Delta y_{N_y-2,N_y-1})$$

$$\begin{aligned}
A^y_{I,j} &= \begin{pmatrix} a^{P_y}_{I,j} & -a^{N_y}_{I,j+1} & 0_{N_y-j_0-2} & -a^{E_y}_{I+1,j} & 0_{N_y-j_0-1} & 0_{(N_y-j_0)(N_x-I_0-2)} \end{pmatrix} \\
A^y_{I,j+1} &= \begin{pmatrix} -a^{S_y}_{I,j} & a^{P_y}_{I,j+1} & -a^{N_y}_{I,j+2} & 0_{N_y-j_0-3} & 0 & -a^{E_y}_{I+1,j+1} & 0_{N_y-j_0-2} & 0_{(N_y-j_0)(N_x-I_0-2)} \end{pmatrix} \\
&\vdots \\
A^y_{I,N_y-1} &= \begin{pmatrix} 0_{N_y-j_0-2} & -a^{S_y}_{I,N_y-2} & a^{P_y}_{I,N_y-1} & 0 & -a^{E_y}_{I+1,j+1} & 0_{N_y-j_0-2} & 0_{(N_y-j_0)(N_x-I_0-2)} \end{pmatrix} \\
A^y_{I+1,j} &= \begin{pmatrix} -a^{W_y}_{I,j} & 0_{N_y-j_0-1} & a^{P_y}_{I+1,j} & -a^{N_y}_{I+1,j+1} & 0_{N_y-j_0-2} & -a^{E_y}_{I+2,j} & 0_{N_y-j_0-1} & 0_{(N_y-j_0)(N_x-i_0-2)} \end{pmatrix} \\
A^y_{I+1,j+1} &= \begin{pmatrix} 0 & -a^{W_y}_{I,j+1} & 0_{N_y-j_0-2} & -a^{S_y}_{I+1,j} & a^{P_y}_{I+1,j+1} & -a^{N_y}_{I+1,j+2} & 0_{N_y-j_0-3} & 0 & -a^{E_y}_{I+2,j+1} & 0_{N_y-j_0-2} \end{pmatrix} \\
&\vdots \\
A^y_{N_x-1,j} &= \begin{pmatrix} 0_{(N_y-j_0)(N_x-I_0-2)} & -a^{W_y}_{N_x-2,j} & 0_{N_y-j_0-1} & a^{P_y}_{N_x-1,j} & -a^{N_y}_{N_x-1,j+1} & 0_{N_y-j_0-2} \end{pmatrix} \\
&\vdots \\
A^y_{N_x-1,N_y-1} &= \begin{pmatrix} 0_{(N_y-j_0)(N_x-I_0-2)} & 0_{N_y-j_0-1} & -a^{W_y}_{N_x-2,N_y-1} & 0_{N_y-j_0-2} & -a^{S_y}_{N_x-1,N_y-2} & a^{P_y}_{N_x-1,N_y-1} \end{pmatrix}
\end{aligned}$$

and

$$A_y = \begin{bmatrix} A^y_{I,j} \\ A^y_{I,j+1} \\ \vdots \\ A^y_{I,N_y-1} \\ A^y_{I+1,j} \\ A^y_{I+1,j+1} \\ \vdots \\ A^y_{N_x-1,j} \\ \vdots \\ A^y_{N_x-1,N_y-1} \end{bmatrix}$$

At last we have

$$\begin{aligned}
B^y_{I,j} &= \begin{pmatrix} -\Delta x_{i-1,i} & \Delta x_{i-1,i} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)(N_x-I_0-1)} \end{pmatrix} \\
B^y_{I,j+1} &= \begin{pmatrix} 0 & -\Delta x_{i-1,i} & \Delta x_{i-1,i} & 0_{N_y-J_0-3} & 0_{(N_y-J_0)(N_x-I_0-1)} \end{pmatrix} \\
&\vdots \\
B^y_{I,N_y-1} &= \begin{pmatrix} 0_{N_y-J_0-2} & -\Delta x_{i-1,i} & \Delta x_{i-1,i} & 0_{(N_y-J_0)(N_x-I_0-1)} \end{pmatrix} \\
B^y_{I+1,j} &= \begin{pmatrix} 0_{N_y-J_0} & -\Delta x_{i,i+1} & \Delta x_{i,i+1} & 0_{N_y-J_0-2} & 0_{(N_y-J_0)(N_x-I_0-2)} \end{pmatrix} \\
B^y_{I+1,j+1} &= \begin{pmatrix} 0_{N_y-J_0} & 0 & -\Delta x_{i,i+1} & \Delta x_{i,i+1} & 0_{N_y-J_0-3} & 0_{(N_y-J_0)(N_x-I_0-2)} \end{pmatrix} \\
&\vdots \\
B^y_{N_x-1,j} &= \begin{pmatrix} 0_{(N_y-J_0)(N_x-I_0-1)} & -\Delta x_{N_x-1,N_x} & \Delta x_{N_x-1,N_x} & 0_{N_y-J_0-2} \end{pmatrix} \\
&\vdots \\
B^y_{N_x-1,N_y-1} &= \begin{pmatrix} 0_{(N_y-J_0)(N_x-I_0-1)} & 0_{N_y-J_0-2} & -\Delta x_{N_x-1,N_x} & \Delta x_{N_x-1,N_x} \end{pmatrix}
\end{aligned}$$

so that we can construct

$$B_y = \begin{bmatrix} B^y_{I,j} \\ B^y_{I,j+1} \\ \vdots \\ B^y_{I,N_y-1} \\ B^y_{I+1,j} \\ B^y_{I+1,j+1} \\ \vdots \\ B^y_{N_x-1,j} \\ \vdots \\ B^y_{N_x-1,N_y-1} \end{bmatrix}$$

The coefficients are defined as:

$$a^{E_y}_{I+1,j} = \max\{-F^{e_y}_{i,j}, D^{e_y}_{i,j} - \frac{F^{e_y}_{i,j}}{2}, 0\}, \qquad a^{W_y}_{I-1,j} = \max\{F^{w_y}_{i-1,j}, D^{w_y}_{i-1,j} + \frac{F^{w_y}_{i-1,j}}{2}, 0\}$$

$$a^{N_y}_{I,j+1} = \max\{-F^{n_y}_{I,J+1}, D^{n_y}_{I,J+1} - \frac{F^{n_y}_{I,J+1}}{2}, 0\}, \quad a^{S_y}_{I,j-1} = \max\{F^{s_y}_{I,J}, D^{s_y}_{I,J} + \frac{F^{a_y}_{I,J}}{2}, 0\}$$

$a^{P_y}_{I,j} = a^{E_y}_{I+1,j} + a^{W_y}_{I-1,j} + a^{N_y}_{I,j+1} + a^{S_y}_{I,j-1} + F^{e_y}_{i,j} - F^{w_y}_{i-1,j} + F^{n_y}_{I,J+1} - F^{s_y}_{I,J}$. With

$$D^{e_y}_{i,j} = \frac{\mu \Delta y_{J,J+1}}{\Delta x_{I,I+1}}, \qquad D^{w_y}_{i-1,j} = \frac{\mu \Delta y_{J,J+1}}{\Delta x_{I-1,I}} \tag{7}$$

$$D^{n_y}_{I,J+1} = \frac{\mu \Delta x_{i-1,i}}{\Delta y_{j,j+1}}, \qquad D^{s_y}_{I,J} = \frac{\mu \Delta x_{i-1,i}}{\Delta y_{j-1,j}} \tag{8}$$

$$F^{e_y}_{i,j} = \frac{\rho \Delta y_{J,J+1}}{2}\left(u_{i,J} + u_{i,J+1}\right), \quad F^{w_y}_{i-1,j} = \frac{\rho \Delta y_{J,J+1}}{2}\left(u_{i-1,J} + u_{i-1,J+1}\right) \tag{9}$$

$$F^{n_y}_{I,J} = \frac{\rho \Delta x_{i-1,i}}{2}\left(v_{I,j+1} + v_{I,j}\right), \quad F^{s_y}_{I,J-1} = \frac{\rho \Delta x_{i-1,i}}{2}\left(v_{I,j-1} + v_{I,j}\right) \tag{10}$$

Continuity equation

In this section we discretise the continuity equation using again the finite volume method. The continuity equation is defined as

$$\rho\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{11}$$

which after integrating over the control volume becomes

$$\rho\Big((u\delta_{y_v})_e - (u\delta_{y_v})_w + (v\delta_{x_u})_n - (v\delta_{x_u})_s\Big) = 0 \tag{12}$$

The general form of the continuity equation

$$\rho\Big((u_{i,J}-u_{i-1,J})\Delta y_{j-1,j}+(v_{I,j}-v_{I,j-1})\Delta x_{i-1,i}\Big)=0 \tag{13}$$

with $I\in[2,N_x-1], i\in[3,N_x-1], J\in[2,N_y-1], j\in[3,N_y]$. We define the following equation:

$$B_u u + B_v v = b^0 \tag{14}$$

With

$$\begin{aligned}
B^u_{i,J} &= \begin{pmatrix}\Delta y_{j-1,j} & 0_{N_y-J^x_0-1} & 0_{(N_y-J^x_0)(N_x-i^x_0-1)}\end{pmatrix}\\
B^u_{i,J+1} &= \begin{pmatrix}0 & \Delta y_{j,j+1} & 0_{N_y-J^x_0-2} & 0_{(N_y-J^x_0)(N_x-i^x_0-1)}\end{pmatrix}\\
&\vdots\\
B^u_{i,N_y-1} &= \begin{pmatrix}0_{N_y-J^x_0-1} & \Delta y_{N_y-1,N_y} & 0_{(N_y-J^x_0)(N_x-i^x_0-1)}\end{pmatrix}\\
B^u_{i+1,J} &= \begin{pmatrix}-\Delta y_{j-1,j} & 0_{N_y-J^x_0-1} & \Delta y_{j-1,j} & 0_{N_y-J^x_0-1} & 0_{(N_y-J^x_0)(N_x-i^x_0-2)}\end{pmatrix}\\
B^u_{i+1,J+1} &= \begin{pmatrix}0 & -\Delta y_{j,j+1} & 0_{N_y-J_0-2} & 0 & \Delta y_{j,j+1} & 0_{N_y-J^x_0-2} & 0_{(N_y-J^x_0)(N_x-i^x_0-2)}\end{pmatrix}\\
&\vdots\\
B^u_{N_x,J} &= \begin{pmatrix}0_{(N_y-J^x_0)(N_x-i^x_0-1)} & -\Delta y_{j-1,j} & 0_{N_y-J^x_0-1}\end{pmatrix}\\
&\vdots\\
B^u_{N_x,N_y-1} &= \begin{pmatrix}0_{(N_y-J^x_0)(N_x-i^x_0-1)} & 0_{N_y-J^x_0-1} & -\Delta y_{N_y-1,N_y}\end{pmatrix}
\end{aligned}$$

and

$$\begin{aligned}
B^v_{I,j} &= \begin{pmatrix}\Delta x_{i-1,i} & 0_{N_y-J_0-1} & 0_{(N_y-j^y_0-1)(N_x-I^y_0)}\end{pmatrix}\\
B^v_{I,j+1} &= \begin{pmatrix}-\Delta x_{i-1,i} & \Delta x_{i-1,i} & 0_{N_y-j^y_0-2} & 0_{(N_y-j^y_0-1)(N_x-I^y_0)}\end{pmatrix}\\
&\vdots\\
B^v_{I,N_y} &= \begin{pmatrix}0_{N_y-j^y_0-1} & -\Delta x_{i-1,i} & 0_{(N_y-j^y_0-1)(N_x-I^y_0)}\end{pmatrix}\\
B^v_{I+1,j} &= \begin{pmatrix}0_{N_y-j^y_0} & -\Delta x_{i,i+1} & 0_{N_y-j^y_0-1} & 0_{(N_y-j^y_0-2)(N_x-I^y_0)}\end{pmatrix}\\
B^v_{I+1,j+1} &= \begin{pmatrix}0_{N_y-j^y_0} & 0 & -\Delta x_{j-1,j} & \Delta x_{j-1,j} & 0_{N_y-j^y_0-3} & 0_{(N_y-j^y_0-2)(N_x-I^y_0)}\end{pmatrix}\\
&\vdots\\
B^v_{N_x-1,j} &= \begin{pmatrix}0_{(N_y-j^y_0-2)(N_x-I^y_0)} & -\Delta x_{j-1,j} & 0_{N_y-j^y_0-1} & \Delta x_{j-1,j} & 0_{N_y-j^y_0-1}\end{pmatrix}\\
&\vdots\\
B^v_{N_x-1,N_y} &= \begin{pmatrix}0_{(N_y-j^y_0-2)(N_x-I^y_0)} & 0_{N_y-j^y_0-1} & -\Delta x_{N_y-1,N_y} & 0_{N_y-j^y_0-1} & \Delta x_{N_y-1,N_y}\end{pmatrix}
\end{aligned}$$

$$b^0 = \begin{pmatrix} \Delta y_{j-1,j} u_{i-1,J} + \Delta x_{i-1,i} v_{I,j-1} \\ \Delta y_{j,j+1} u_{i-1,J+1} \\ \vdots \\ \Delta y_{N_y-1,N_y} u_{i-1,N_y-1} + \Delta x_{N_y-1,N_y} v_{I,N_y} \\ \Delta x_{i,i-1} v_{I+1,j-1} \\ 0 \\ \vdots \\ -\Delta y_{j-1,j} u_{N_x,J} - \Delta x_{N_x-1,N_x} v_{N_y,j-1} \\ \vdots \\ -\Delta y_{N_y-1,N_y} u_{N_x,N_y-1} - \Delta x_{N_x-1,N_x} v_{N_y-1,N_y} \end{pmatrix} \tag{15}$$

General Formulation

We can write the above three equations in one equation as follows:

$$\begin{bmatrix} A_x & 0 & B_x \\ 0 & A_y & B_y \\ B_u & B_v & 0 \end{bmatrix} \begin{pmatrix} u \\ v \\ p \end{pmatrix} = \begin{bmatrix} u^0 + V_x \left(S_x - \rho \frac{\partial u}{\partial t}\right) \\ v^0 + V_y \left(S_y - \rho \frac{\partial v}{\partial t}\right) \\ b^0 \end{bmatrix} \tag{16}$$

Which is equivalent to

$$\begin{bmatrix} A_x & 0 & B_x \\ 0 & A_y & B_y \\ B_x^T & B_y^T & 0 \end{bmatrix} \begin{pmatrix} u \\ v \\ p \end{pmatrix} = \begin{bmatrix} u^0 + V_x \left(S_x - \rho \frac{\partial u}{\partial t}\right) \\ v^0 + V_y \left(S_y - \rho \frac{\partial v}{\partial t}\right) \\ -b^0 \end{bmatrix} \tag{17}$$

where we multiplied the continuity equation with -1.

2 Temporal Discretisation

In this section we will temporal discretise the above derived equations. We will use the implicit scheme for this as can be found in **?**[Versteeg: An introduction to CFD, page 257].

u-momentum equation

We had found the following spatially discretised formulation for the u-momentum equation

$$\left(S^x_{i,J} - \rho \frac{\partial u_{i,J}}{\partial t}\right) \overbrace{\underbrace{(x_{I+1,J} - x_{I,J})}_{\Delta x_{I,I+1}} \underbrace{(y_{i,j} - y_{i,j-1})}_{\Delta y_{j-1,j}}}^{\Delta V_u} = a^{P_x}_{i,J} u_{i,J} - a^{E_x}_{i+1,J} u_{i+1,J} - a^{W_x}_{i-1,J} u_{i-1,J} - \cdots$$
$$\cdots - a^{N_x}_{i,J+1} u_{i,J+1} - a^{S_x}_{i,J-1} u_{i,J-1} + (p_{I+1,J} - p_{I,J}) \underbrace{(y_{i,j} - y_{i,j-1})}_{\Delta y_{j-1,j}}$$

The corresponding coefficients are defined in Equation 6. It can be shown that after temporally discretising the equations we obtain

$$S_u = a^{P_x}_{i,J} u_{i,J} - a^{P^0_x}_{i,J} u^0_{i,J} - a^{E_x}_{i+1,J} u_{i+1,J} - a^{W_x}_{i-1,J} u_{i-1,J} - \cdots$$
$$\cdots - a^{N_x}_{i,J+1} u_{i,J+1} - a^{S_x}_{i,J-1} u_{i,J-1} + (p_{I+1,J} - p_{I,J}) \Delta y_{j-1,j}$$

with $a^{P_x}_{i,J} = a^{E_x}_{i+1,J} + a^{W_x}_{i-1,J} + a^{N_x}_{i,J+1} + a^{S_x}_{i,J-1} + F^{e_x}_{I+1,J} - F^{w_x}_{I,J} + F^{n_x}_{i,j} - F^{s_x}_{i,j-1} + a^{P^0_x}_{i,J} - S^{P_x}_{i,J}$ and

$$a^{P^0_x}_{i,J} = \frac{\rho \Delta V_u}{\Delta t}, \qquad S_u = S^x_{i,J} \Delta V_u - S^{P_x}_{i,J} u_{i,J} \tag{18}$$

v-momentum equation

$$\left(S^y_{I,j} - \rho\frac{\partial v_{I,j}}{\partial t}\right)\overbrace{\underbrace{(x_{i,j}-x_{i-1,j})}_{\Delta x_{i-1,i}}\underbrace{(y_{I,J+1}-y_{I,J})}_{\Delta y_{J,J+1}}}^{\Delta V_v} = a^{P_y}_{I,j}v_{I,j} - a^{E_y}_{I+1,j}v_{I+1,j} - a^{W_y}_{I-1,j}v_{I-1,j} - \cdots$$
$$\cdots - a^{N_y}_{I,j+1}v_{I,j+1} - a^{S_y}_{I,j-1}v_{I,j-1} + (p_{I,J+1}-p_{I,J})\underbrace{(x_{i,j}-x_{i-1,j})}_{\Delta x_{i-1,i}}$$

The corresponding coefficients are defined in Equation 10. It can be shown that after temporally discretising the equations we obtain

$$S_v = a^{P_y}_{I,j}v_{I,j} - a^{P^0_y}_{I,j}v^0_{I,j} - a^{E_y}_{I+1,j}v_{I+1,j} - a^{W_y}_{I-1,j}v_{I-1,j} - \cdots$$
$$\cdots - a^{N_y}_{I,j+1}v_{I,j+1} - a^{S_y}_{I,j-1}v_{I,j-1} + (p_{I,J+1}-p_{I,J})\,\Delta x_{i-1,i}$$

with $a^{P_y}_{I,j} = a^{E_y}_{I+1,j} + a^{W_y}_{I-1,j} + a^{N_y}_{I,j+1} + a^{S_y}_{I,j-1} + F^{e_y}_{i,j} - F^{w_y}_{i-1,j} + F^{n_y}_{I,J+1} - F^{s_y}_{I,J} + a^{P^0_y}_{I,j} - S^{P_y}_{I,j}$ and

$$a^{P^0_v}_{I,j} = \frac{\rho\Delta V_v}{\Delta t}, \qquad S_v = S^y_{I,j}\Delta V_v - S^{P_y}_{I,j}v_{I,j} \tag{19}$$

General Formulation

The general formulation then becomes:

$$\begin{bmatrix} A_x & 0 & B_x \\ 0 & A_y & B_y \\ B_x^T & B_y^T & 0 \end{bmatrix}\begin{pmatrix} u \\ v \\ p \end{pmatrix} = \begin{bmatrix} u^0 + S_u \\ v^0 + S_v \\ -b^0 \end{bmatrix} \tag{20}$$

And the matrices change according to the temporal discretisation given in the previous subsections.

3 Boundary Conditions

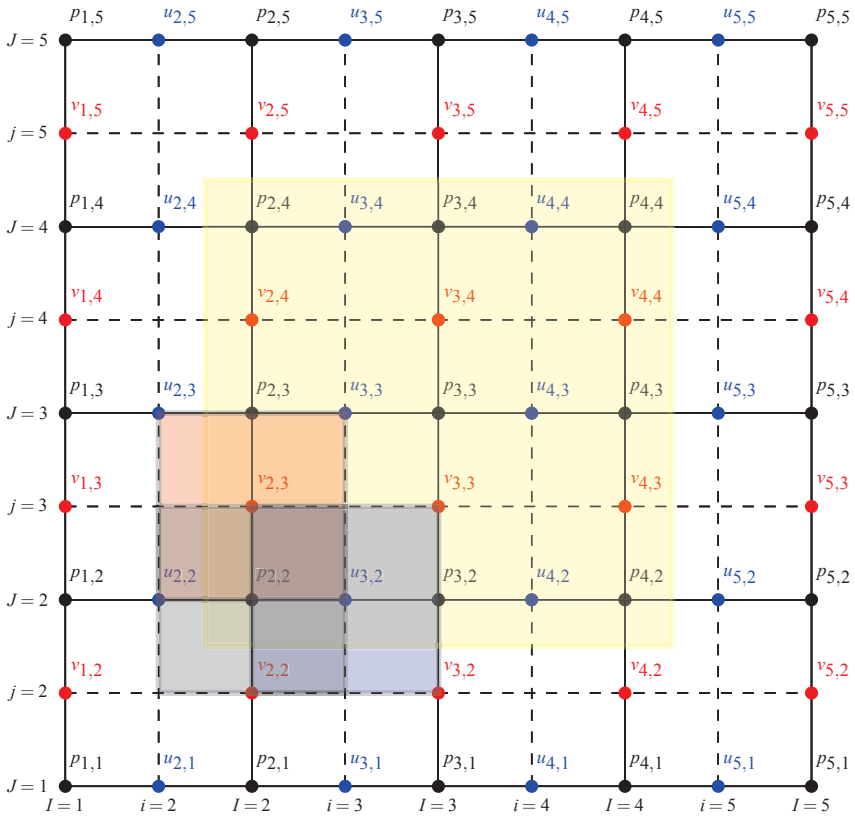


Figure 2: Example reference frame used to discretise the equations.

Example

In this section we provide the reader with an example in order to illustrate the structure of the matrices. When defining the following equations

$$A_x u + B_x p =$$

We in this example have

$$
\begin{aligned}
u &= \begin{pmatrix} u_{3,2} & u_{3,3} & u_{3,4} & u_{4,2} & u_{4,3} & u_{4,4} \end{pmatrix}^T \\
u_0 &= \begin{pmatrix} a^{W_x}u_{2,2} + a^{S_x}u_{3,1} & a^{W_x}u_{2,3} & a^{W_x}u_{2,4} + a^{N_x}u_{3,5} & a^{S_x}u_{4,1} + a^{E_x}u_{5,2} & a^{E_x}u_{5,3} & a^{N_x}u_{4,5} + a^{E_x}u_{5,4} \end{pmatrix}^T \\
v &= \begin{pmatrix} v_{2,3} & v_{3,4} & v_{3,3} & v_{3,4} & v_{4,3} & v_{4,4} \end{pmatrix}^T \\
v_0 &= \begin{pmatrix} a^{W_y}v_{1,3} + a^{S_y}v_{2,2} & a^{W_y}v_{1,4} + a^{N_y}v_{2,5} & a^{S_y}v_{3,2} & a^{N_y}v_{3,5} & a^{S_y}u_{4,2} + a^{E_y}v_{5,3} & a^{N_y}v_{4,5} + a^{E_y}v_{5,4} \end{pmatrix}^T \\
p &= \begin{pmatrix} p_{2,2} & p_{2,3} & p_{2,4} & p_{3,2} & p_{3,3} & p_{3,4} & p_{4,2} & p_{4,3} & p_{4,4} \end{pmatrix}^T \\
V_x &= \texttt{diag}(\Delta x_{2,3}\Delta y_{2,3}, \Delta x_{2,3}\Delta y_{3,4}, \Delta x_{2,3}\Delta y_{4,5}, \Delta x_{3,4}\Delta y_{2,3}, \Delta x_{3,4}\Delta y_{3,4}, \Delta x_{3,4}\Delta y_{4,5}) \\
V_y &= \texttt{diag}(\Delta x_{2,3}\Delta y_{2,3}, \Delta x_{2,3}\Delta y_{3,4}, \Delta x_{3,4}\Delta y_{2,3}, \Delta x_{3,4}\Delta y_{3,4}, \Delta x_{4,5}\Delta y_{2,3}, \Delta x_{4,5}\Delta y_{3,4})
\end{aligned}
$$

and

$$
A_x = \begin{pmatrix}
a^{P_x}_{3,2} & -a^{N_x}_{3,3} & 0 & -a^{E_x}_{4,2} & 0 & 0 \\
-a^{S_x}_{3,2} & a^{P_x}_{3,3} & -a^{N_x}_{3,4} & 0 & -a^{E_x}_{4,3} & 0 \\
0 & -a^{S_x}_{3,3} & a^{P_x}_{3,4} & 0 & 0 & -a^{E_x}_{4,4} \\
-a^{W_x}_{3,2} & 0 & 0 & a^{P_x}_{4,2} & -a^{N_x}_{4,3} & 0 \\
0 & -a^{W_x}_{3,3} & 0 & a^{S_x}_{4,2} & a^{P_x}_{4,3} & -a^{N_x}_{4,4} \\
0 & 0 & -a^{W_x}_{3,4} & 0 & -a^{S_x}_{4,3} & a^{P_x}_{4,4}
\end{pmatrix} \tag{21}
$$

$$
B_x = \begin{pmatrix}
-\Delta y_{2,3} & 0 & 0 & \Delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
0 & -\Delta y_{3,4} & 0 & 0 & \Delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & 0 & -\Delta y_{4,5} & 0 & 0 & \Delta y_{4,5} & 0 & 0 & 0 \\
0 & 0 & 0 & -\Delta y_{2,3} & 0 & 0 & \Delta y_{2,3} & 0 & 0 \\
0 & 0 & 0 & 0 & -\Delta y_{3,4} & 0 & 0 & \Delta y_{3,4} & 0 \\
0 & 0 & 0 & 0 & 0 & -\Delta y_{4,5} & 0 & 0 & \Delta y_{4,5}
\end{pmatrix} \tag{22}
$$

$$
B_u = \begin{pmatrix}
\Delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
0 & \Delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & 0 & \Delta y_{4,5} & 0 & 0 & 0 \\
-\Delta y_{2,3} & 0 & 0 & \Delta y_{2,3} & 0 & 0 \\
0 & -\Delta y_{3,4} & 0 & 0 & \Delta y_{3,4} & 0 \\
0 & 0 & -\Delta y_{4,5} & 0 & 0 & \Delta y_{4,5} \\
0 & 0 & 0 & -\Delta y_{2,3} & 0 & 0 \\
0 & 0 & 0 & 0 & -\Delta y_{3,4} & 0 \\
0 & 0 & 0 & 0 & 0 & -\Delta y_{4,5}
\end{pmatrix} \tag{23}
$$

$$A_y = \begin{pmatrix} a_{2,3}^{P_y} & -a_{2,4}^{N_y} & -a_{3,3}^{E_y} & 0 & 0 & 0 \\ -a_{2,3}^{S_y} & a_{2,4}^{P_y} & 0 & -a_{3,4}^{E_y} & 0 & 0 \\ -a_{2,3}^{W_y} & 0 & a_{3,3}^{P_y} & -a_{3,4}^{N_y} & -a_{4,3}^{E_y} & 0 \\ 0 & -a_{2,4}^{W_y} & -a_{3,3}^{S_y} & a_{3,4}^{P_y} & 0 & -a_{4,4}^{E_y} \\ 0 & 0 & -a_{3,3}^{W_y} & 0 & a_{4,3}^{P_y} & -a_{4,4}^{N_y} \\ 0 & 0 & 0 & -a_{3,4}^{W_y} & -a_{4,3}^{S_y} & a_{4,4}^{P_x} \end{pmatrix} \tag{24}$$

$$B_y = \begin{pmatrix} -\Delta x_{2,3} & \Delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta x_{2,3} & \Delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\Delta x_{3,4} & \Delta x_{3,4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\Delta x_{3,4} & \Delta x_{3,4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\Delta x_{4,5} & \Delta x_{4,5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\Delta x_{4,5} & \Delta x_{4,5} \end{pmatrix} \tag{25}$$

$$B_v = \begin{pmatrix} \Delta x_{2,3} & 0 & 0 & 0 & 0 & 0 \\ -\Delta x_{2,3} & \Delta x_{2,3} & 0 & 0 & 0 & 0 \\ 0 & -\Delta x_{2,3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \Delta x_{3,4} & 0 & 0 & 0 \\ 0 & 0 & -\Delta x_{3,4} & \Delta x_{3,4} & 0 & 0 \\ 0 & 0 & 0 & -\Delta x_{3,4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta x_{4,5} & 0 \\ 0 & 0 & 0 & 0 & -\Delta x_{4,5} & \Delta x_{4,5} \\ 0 & 0 & 0 & 0 & 0 & -\Delta x_{4,5} \end{pmatrix} \tag{26}$$

$$b^0 = \begin{pmatrix} \Delta y_{2,3} u_{2,2} + \Delta x_{2,3} v_{2,2} \\ \Delta y_{3,4} u_{2,3} \\ \Delta y_{4,5} u_{2,4} + \Delta x_{2,3} v_{2,5} \\ \Delta x_{3,4} v_{3,2} \\ 0 \\ -\Delta x_{3,4} v_{3,5} \\ -\Delta y_{2,3} u_{5,2} + \Delta x_{4,5} v_{4,2} \\ -\Delta y_{3,4} u_{5,3} \\ -\Delta y_{4,5} u_{5,4} - \Delta x_{4,5} v_{4,5} \end{pmatrix} \tag{27}$$

4 General formulation

$$\begin{bmatrix} A_x & 0 & B_x \\ 0 & A_y & B_y \\ B_u & B_v & 0 \end{bmatrix} \begin{pmatrix} u \\ v \\ p \end{pmatrix} = \begin{bmatrix} u^0 + V_x \left(S_x - \rho \frac{\partial u}{\partial t} \right) \\ v^0 + V_y \left(S_y - \rho \frac{\partial v}{\partial t} \right) \\ b^0 \end{bmatrix} \tag{28}$$

Which is equivalent to

$$\begin{bmatrix} A_x & 0 & B_x \\ 0 & A_y & B_y \\ B_x^T & B_y^T & 0 \end{bmatrix} \begin{pmatrix} u \\ v \\ p \end{pmatrix} = \begin{bmatrix} u^0 + V_x \left(S_x - \rho \frac{\partial u}{\partial t} \right) \\ v^0 + V_y \left(S_y - \rho \frac{\partial v}{\partial t} \right) \\ -b^0 \end{bmatrix} \tag{29}$$

where we multiplied the continuity equation with -1.